Homework 5
APPM 4490/5490 Theory of Machine Learning, Spring 2026

Due date: Wednesday, Feb 18 '26, before noon, via paper or via Gradescope

Instructor: Prof. Becker
Revision date: 2/10/2026

Theme: VC dimension

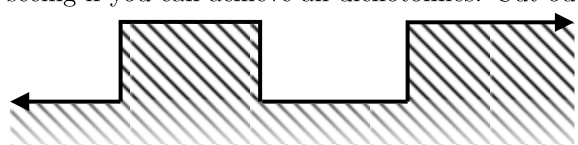
Instructions Collaboration with your fellow students is OK and in fact recommended, although direct copying is not allowed. The internet is allowed for basic tasks (e.g., looking up definitions on wikipedia) but it is not permissible to search for proofs or to *post* requests for help on forums such as `http://math.stackexchange.com/` or to look at solution manuals; see also the class AI policy on our website. Please write down the names of the students that you worked with.

An arbitrary subset of these questions will be graded.

Reading You are responsible for reading chapter 6 of "Understanding Machine Learning" by Shai Shalev-Shwartz and Shai Ben-David (2014, Cambridge University Press). Note: you can buy the book for about $45 on Amazon (or less for an e-book), and the authors host a free PDF copy on their website (but note that this PDF has different page numbers).

Note For VC dimension calculations, you should have a convincing argument, but you don't need to spend a very long time writing out a rigorous proof (e.g., when appropriate, a "proof by pictures" or a "I checked all possibilities and it works" may suffice). When in doubt, ask the instructor at office hours.

Problem 1: Exercise 3.15 in Mohri et al: let $I_\alpha = [\alpha, \alpha+1] \cup [\alpha+2, \infty)$, and the classifier $h_\alpha(x) = 1_{I_\alpha}(x)$ the indicator function associated with this set. Let $\mathcal{H}$ be the set of all such classifiers h_α. What is the VC dimension of $\mathcal{H}$?

Hint: For fun (and because it is helpful), for a set C, you can check if it is shattered by this $\mathcal{H}$ by sliding a template over the points and seeing if you can achieve all dichotomies. Cut out a paper template in the shape of the h, i.e., 

Problem 2: Exercise 6.2a in Shalev-Shwartz and Ben-David: Let the domain $\mathcal{X}$ be a finite set with $|\mathcal{X}| = n$, and fix any integer $k \in [0, n]$. Let $\mathcal{H}$ be the set of all binary functions on $\mathcal{X}$ that give an output of 1 for exactly k of the inputs. What is the VC dimension of $\mathcal{H}$?

Note: Exercise 6.2b is not required, but if you want to do it for extra practice, define $\mathcal{H}$ to be the set of all binary functions that gives an output of 1 for at most k of the inputs (in the textbook, ignore the confusing "or" statement in the definition of $\mathcal{H}$).

Problem 3: [**Reading**] Skim Real numbers, data science and chaos: How to fit any dataset with a single parameter (arXiv: 1904.12320) by Laurent Boué; you don't need to understand the technical construction (unless you're curious). Write a short response (a few sentences) about the high-level issues the paper raises.

Like our $\lceil \sin(\theta x) \rceil$ example, this is a case where the VC dimension is *not* equal to the number of parameters. One issue with parameter counting is that a real-valued parameter isn't easy to

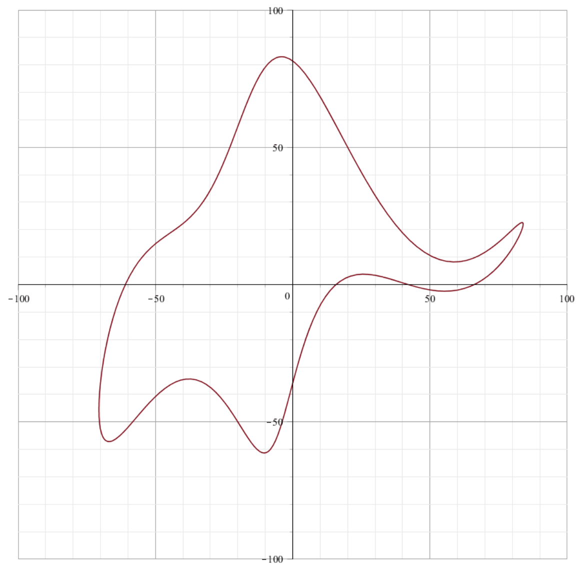


Figure 1: Von Neumann's elephant; CC BY-SA 4.0 license, created by user Kelam, unmodified.

quantify in an information theoretic way (in contrast to discrete data, like a bit), related to the fact that $\mathbb{R}$ and $\mathbb{R}^2$ have the same cardinality. You could try to treat a real valued parameter θ as the limit of rational approximations, but that breaks down with the system is chaotic. The images in Boué's paper are modern versions of Von Neumann's elephant. Quoting from wikipedia,

> In a 2004 article in the journal Nature, Freeman Dyson recounts his meeting with Fermi in 1953. Fermi, while discussing a novel theory Dyson was proposing, offered the harsh critique, "There are two ways of doing calculations in theoretical physics. One way ... is to have a clear physical picture of the process that you are calculating. The other way is to have a precise and self-consistent mathematical formalism. You have neither." Dyson countered by stating that his theory matched Fermi's own data. Fermi asked about a number of arbitrary parameters Dyson used and, upon learning that there were four of them, quoted his friend von Neumann,
>
>> "With four parameters I can fit an elephant, and with five I can make him wiggle his trunk."
>
> By this he meant that the Dyson's simulations relied on too many free parameters, presupposing an overfitting phenomenon. "Stunned" Dyson agreed with the argument, finished the set of articles in order for his students to get their names into the research journals, and switched to another field of study.

The figure above is Von Neumann's elephant, from Wikipedia.